

DROIDBRIDGE

ADB-Powered Android Device Management Tool

Project Requirements & Feature Specification

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Open Source Project

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1. Project Overview

DroidBridge is an open-source, cross-platform desktop tool for managing Android devices via ADB (Android Debug Bridge). It provides significantly faster file transfers than standard MTP, intelligent media analysis and organization, a complete WhatsApp backup and cleanup toolkit, app management, and rich report generation — all without requiring internet access or cloud services.

The tool was born from a real need: Android users face painfully slow MTP transfers, no smart file discovery, and no native tools for managing large WhatsApp media archives accumulated over years. DroidBridge solves all of these problems in a single, privacy-first, offline application.

Core Value Proposition: 3–5x faster than MTP | Smart media analysis | Complete WhatsApp toolkit | 100% offline & private

2. Goals & Design Principles

Primary Goals

- Replace slow MTP with ADB-powered transfers that are 3–5x faster
- Provide complete WhatsApp media analysis, backup, organization, and cleanup
- Enable intelligent storage analysis and space recovery on Android devices
- Work entirely offline with no cloud dependency or internet requirement
- Be accessible via both CLI (for power users) and GUI (for general users)
- Generate detailed human-readable reports for all operations

Design Principles

Principle	Description
Safety First	No destructive operation runs without explicit confirmation. Every deletion shows a preview first.
Verified Backup	All backups are verified with file count and size checks before any phone deletion.
Resume Support	All transfers and operations can be interrupted and resumed without starting over.
Offline & Private	No internet required. No telemetry. No cloud. All data stays local.
Cross-Platform	Works on Windows, Linux, and macOS with the same feature set.
ADB Bundled	ADB is bundled with the tool — no separate Android SDK installation required.
Organized Output	All backups are automatically organized into intuitive folder structures.
Detailed Logging	Every session is logged with timestamps, counts, sizes, and error reports.

3. Technology Stack

Component	Technology	Reason
Core Language	Python 3.10+	Cross-platform, rich ecosystem, fast prototyping
CLI Interface	Click / argparse	Clean argument parsing, help generation
GUI Framework	PyQt6 or Tkinter	Cross-platform desktop UI, no browser required
ADB Communication	subprocess + adb binary	Direct ADB commands, no Java dependency
ADB Binary	Bundled platform-tools	No user installation required
File Analysis	os, pathlib, re, mimetypes	Built-in, no dependencies
Report Generation	Jinja2 + HTML/PDF	Rich formatted reports
Data Storage	SQLite	Local session and backup metadata
Packaging	PyInstaller	Single executable per platform
Version Control	Git + GitHub	Open source collaboration

Note: The tool bundles ADB platform-tools for Windows, Linux, and macOS. Users do not need to install Android SDK separately.

3a. Platform Strategy

DroidBridge is built as a single Python codebase shared across all platforms — not separate codebases per OS. Platform-specific differences (ADB binary, sleep inhibitor, paths) are isolated behind a small platform abstraction layer.

Single Codebase, Per-Platform Releases

- One Python codebase for core logic, CLI, and GUI
- Platform-specific code isolated in `core/platform/` (`windows.py`, `linux.py`, `macos.py`)
- Bundled ADB binaries: `platform-tools-windows`, `platform-tools-linux`, `platform-tools-darwin`
- Sleep inhibitor implementations per OS:
 - Linux: `systemd-inhibit`
 - Windows: `SetThreadExecutionState` (ctypes)
 - macOS: `caffeinate` subprocess
- Build & release separately per platform via `PyInstaller` (one executable per OS), but from the same source

Why Not Separate Codebases

- WhatsApp toolkit logic (the core value) is 100% identical across platforms
- Maintaining 3 codebases triples bug-fix effort for marginal gain
- ADB commands themselves are identical across platforms — only the binary path and OS-level helpers differ

Development Order by Platform

Stage	Primary Platform	Reason
Phase 1–3 (CLI core)	Linux (Ubuntu)	Development environment, fastest iteration, matches original sessions
Phase 4 (Polish)	Linux + Windows	Add Windows ADB path handling and sleep inhibitor
Phase 5 (GUI)	Linux + Windows + macOS	GUI framework (PyQt6) is cross-platform from day one
Release	All three	<code>PyInstaller</code> builds produced for Windows, Linux, macOS from same tag

Recommendation: Build and validate on Linux first (matches your existing environment and session history), then add Windows/macOS platform adapters in

Phase 4. The CLI works on all 3 platforms before GUI work begins.

Module 1: Device Manager

Handles device detection, connection management, and device information display.

1.1 Device Detection & Connection

- Auto-detect connected Android devices via ADB on startup
- Display connection status with clear indicators (connected / unauthorized / offline)
- Guide user through USB debugging setup if not enabled
- Support multiple simultaneously connected devices with device selector
- Reconnect automatically if device disconnects and reconnects
- Show device serial number, model name, Android version

1.2 Device Information Display

- Device model, manufacturer, Android version, build number
- Storage breakdown: total, used, free (internal storage)
- Battery level and charging status
- USB connection type (USB 2.0 / 3.0 / 3.1)
- Estimated transfer speed based on connection type

1.3 Storage Overview

- Visual storage bar showing used vs free
- Breakdown by category: Apps, Images, Videos, Audio, Documents, System
- Top space-consuming apps list
- WhatsApp-specific storage summary

1.4 Connection Health

- ADB daemon status check
- Auto-restart ADB daemon if needed
- USB mode detection and guidance (must be File Transfer / MTP)
- Sleep prevention during active operations

Module 2: File Browser

Browse, search, and preview files on the connected Android device without copying them first.

2.1 Directory Navigation

- Tree-view navigation of phone filesystem
- Show file count and total size per folder
- Hidden file toggle (show/hide .nomedia, etc.)
- Breadcrumb navigation path
- Bookmarks for frequently accessed folders
- Quick-jump to common locations (WhatsApp, DCIM, Downloads, etc.)

2.2 File Listing & Filtering

- Sort by: name, date modified, size, file type
- Filter by: file extension, date range, size range
- Show: filename, size, date modified, extension
- Multi-select files and folders
- Select all / deselect all
- Invert selection

2.3 File Preview

- Inline image preview (JPEG, PNG, HEIC, GIF, WebP)
- Video thumbnail preview
- Audio file metadata (duration, bitrate)
- Document type icon with file size
- Preview without copying file to laptop

2.4 File Operations (via right-click menu)

- Copy to laptop (single or batch)
- Delete from phone (with confirmation)
- Rename file on phone
- Show full file path
- Copy file path to clipboard

Module 3: Smart Transfer Engine

ADB-powered file transfer that is 3–5x faster than standard MTP, with resume support, progress tracking, and conflict resolution.

3.1 Transfer Modes

- Phone → Laptop (pull)
- Laptop → Phone (push)
- Batch transfer: multiple files or entire folders
- Selective transfer: transfer only files matching filter criteria
- Mirror mode: sync phone folder to laptop folder

3.2 Performance

- ADB batch commands (500 files per ADB call) for 10–20x speed vs one-by-one
- Automatic USB sleep prevention during transfer
- Transfer speed display (MB/s)
- Estimated time remaining
- Progress bar with file count and bytes transferred

3.3 Resume & Reliability

- Detect interrupted transfers and offer resume
- Skip already-transferred files (by name + size comparison)
- Retry failed transfers automatically (configurable retry count)
- Transfer log saved to disk for post-session review

3.4 Conflict Resolution

- Skip: keep existing file, don't overwrite
- Overwrite: replace existing file
- Rename: add suffix to new file (_dup, _1, _2, etc.)
- Ask each time: prompt user per conflict
- Remember choice for session

3.5 Post-Transfer Verification

- File count verification: source vs destination
- Total size verification
- Generate transfer report with success/failure summary

Module 4: WhatsApp Toolkit

A comprehensive toolkit for WhatsApp media management built from real-world experience. Covers analysis, backup, organization, cleanup, and database preservation.

Important: This module supports both WhatsApp (com.whatsapp) and WhatsApp Business (com.whatsapp.w4b) — selectable at runtime.

4.1 Media Analysis & Reporting

- Scan all WhatsApp media folders on the phone
- Detect WhatsApp media path automatically (varies by Android version):
 - /sdcard/Android/media/com.whatsapp/WhatsApp/Media/ (Android 10+)
 - /sdcard/WhatsApp/Media/ (older versions)
- Generate report broken down by:
 - Folder type (Voice Notes, Images, Video, Documents, Stickers, Audio, Gifs, etc.)
 - Section (Received, Sent, Private)
 - Year and month
 - File type / extension
 - File count and actual size per group
- Pre vs post user-defined cutoff date comparison
- Identify orphaned media: files with no linked chat in database
- Detect files with missing or incorrect extensions using MIME type detection
- Detect files with malformed dates (student IDs, serial numbers mistaken as dates)
- Save analysis reports to disk (TXT and HTML formats)

4.2 Folder Coverage

Folder	Content	Notes
WhatsApp Voice Notes	Voice messages (PTT)	Organized by numbered subfolders on newer Android
WhatsApp Images	Received/Sent/Private images	JPEG, PNG, HEIC, WebP
WhatsApp Video	Received/Sent/Private videos	MP4, MOV, MKV, AVI
WhatsApp Audio	Audio files shared in chats	MP3, AAC, M4A, OGG
WhatsApp Documents	All document types	PDF, DOCX, PPTX, ZIP, RAR, source code, etc.
WhatsApp Stickers	Received stickers	Backed up — NOT included in Google Drive

Folder	Content	Notes
		backup
WhatsApp Sticker Packs	Installed sticker packs	From official WhatsApp store
WhatsApp Backup Excluded Stickers	Third-party sticker packs	Excluded from all backups — manual only
WhatsApp Animated Gifs	GIF files	Received in chats
WhatsApp Profile Photos	Contact profile pictures	Cached profile photos
WhatsApp Video Notes	Short video messages	Circular video bubbles
WallPaper	Chat wallpapers	Custom wallpapers set by user
Databases	msgstore.db and others	Active message database — critical
Backups	Local encrypted backups	.crypt12 / .crypt15 files

4.3 Backup Features

- One-click full WhatsApp backup to selected destination
- Selective backup by media type (e.g. only images + voice notes)
- Direct-to-external-drive backup (bypasses laptop storage)
- Resume interrupted backups using rsync-style logic
- Post-backup verification: file count and size match check
- Backup report saved with timestamp, counts, and sizes
- Backup session logs: all scripts, reports, and summaries preserved
- Prevent laptop sleep during backup using system inhibit

4.4 Backup Organization (Optional)

After pulling files, the user can optionally choose to organize them into clean folder structures. Organization is NOT automatic — it is a separate step the user explicitly triggers after backup is complete.

Organization is optional. Raw backup (flat folder structure) is always the default. The user chooses whether to organize after backup is verified.

Organization options:

- Organize now — run organization immediately after backup
- Organize later — keep raw backup, run organization as a separate step anytime
- Skip organization — keep raw flat folder structure permanently
- Custom organization — user defines folder structure rules

When organization is chosen, the following structures are applied:

Voice Notes:

```
WhatsApp_Voice_Notes_Organized/ 2017/ 2018/ 2019/ ... 2026/ 01-Jan/ 02-Feb/ ... 12-Dec/ PTT-20210615-WA0001.opus Unknown/ ← files with no date pattern
```

Images & Video:

```
WhatsApp_Images_Organized/ Received/ 2020/ 01-Jan/ ... 12-Dec/ Sent/ 2020/ 01-Jan/ ... 12-Dec/ Private/ 2024/ ...
```

Documents:

```
WhatsApp_Documents_Organized/ Received/ PDFs/ 2021/ 01-Jan/ ... Archives/ ... Images/ ... Videos/ ... Source_Code/ Original_Files/ Engineering/ Original_Files/ Presentations/ ... Word_Documents/ ... Spreadsheets/ ... Executables/ ... Other/ ... Sent/ (same structure)
```

4.5 File Fixing & Cleanup

- Detect files with no extension using MIME type fingerprinting
- Automatically rename files with correct extension (e.g. DOC-20220310-WA0001. → .pdf)
- Fix double-dot filenames (e.g. Sheet..pdf → Sheet.pdf)
- Detect and fix files placed in wrong year folders due to numbers in filename
- Move files with original names (no WhatsApp date) to Original_Files subfolder
- Validate year range (2009–2030) to reject false date matches
- Validate month range (01–12) to reject false date matches

4.6 Cleanup & Deletion

- Delete media by date cutoff (e.g. delete everything before September 2024)
- Pre-deletion preview showing exact file count and estimated size by year
- Breakdown of what will be deleted vs kept by folder and file type
- Keep/delete by file type (e.g. keep images and audio, delete documents and archives)
- Batch deletion using 500 files per ADB command (10–20x faster than one-by-one)
- Mandatory confirmation step: user must type 'YES DELETE' to proceed
- Deletion log saved with count of deleted vs failed files
- Post-deletion rescan to verify cleanup was successful
- Resume interrupted deletion: rescan phone and build new delete list from remaining files
- Never delete without verified backup existing first

4.7 Pre/Post Cutoff Comparison

- User defines a cutoff date (default: September 2024)
- Report shows for each folder/type:
 - Pre-cutoff: file count and size
 - Post-cutoff: file count and size

- No-date files: count and size
- Total deletable size estimate
- Status tracking: Done / Partial / Pending per month after deletion

4.8 Database & System Files Backup

- Backup Databases/ folder (msgstore.db, wa.db, axolotl.db, etc.)
- Backup Backups/ folder (local encrypted .crypt files)
- Backup accounts/ folder
- Warn user that msgstore.db contains active chats — critical to preserve
- Note encryption status of database files

Module 5: Storage Analyzer

Analyze phone storage in depth to identify what is taking space and find opportunities for cleanup.

5.1 Full Storage Breakdown

- Total storage, used, and free space
- Breakdown by Android category: Apps (installed + data + cache), Images, Videos, Audio, Documents, Downloads, System
- Visual bar chart of storage usage
- Compare against phone total capacity

5.2 App Storage Analysis

- List all installed apps sorted by total size (installed + data + cache)
- Show per-app breakdown: APK size, data size, cache size
- Identify top 20 space consumers
- Flag apps with unusually large cache (potential cleanup targets)
- Distinguish system apps from user-installed apps

5.3 Media Analysis

- Scan all media across entire phone (not just WhatsApp)
- Breakdown by media type: photos, videos, audio, documents
- Identify largest individual files
- Find files older than a specified date
- Find duplicate files by name and size

5.4 Large File Finder

- Find all files above a configurable size threshold (default: 50 MB)
- Sort by size descending
- Show file path, type, and size
- Option to copy or delete found files

5.5 Cleanup Suggestions

- App cache that can be safely cleared
- APK installer files in Downloads
- Thumbnail cache folders
- Old log files and crash reports
- Estimated space recoverable per suggestion

Module 6: Backup Manager

Manage full phone backups with profiles, scheduling, verification, and restore support.

6.1 Backup Profiles

- Create named backup profiles (e.g. 'WhatsApp Only', 'Full Media', 'Documents')
- Define which folders to include/exclude per profile
- Define destination path per profile
- Save and reuse profiles across sessions

6.2 Backup Execution

- Run full or selective backup based on profile
- Direct-to-external-drive support
- Incremental backup: only copy new/changed files since last backup
- Prevent laptop sleep during backup
- Real-time progress with speed, ETA, and file count

6.3 Backup Verification

- Post-backup file count check: source vs destination
- Post-backup size check: source vs destination
- Flag any files that failed to transfer
- Generate verification report
- Block phone-side deletion until backup is verified

6.4 Backup History

- Log each backup with: timestamp, profile used, file count, total size, duration, destination
- Browse backup history in UI
- Compare backups across sessions
- Detect if backup is outdated (last backup > N days ago)

6.5 Restore

- Restore files from backup to phone
- Selective restore: choose specific folders or date ranges
- WhatsApp restore: push organized backup back to correct phone paths
- Conflict resolution during restore (skip/overwrite/rename)

Module 7: Search & Discovery

Instantly search and discover files across the entire phone filesystem.

7.1 Search Capabilities

- Search by filename (partial match, wildcard, regex)
- Search by file extension or MIME type
- Search by date range (modified date from filename or filesystem)
- Search by size range (e.g. files between 10 MB and 100 MB)
- Scope search to specific folder or entire phone
- Combine multiple filters in a single search

7.2 Search Results

- Show results with: full path, size, date, extension
- Sort results by any column
- Paginate large result sets
- Select results for batch operations (copy, delete)
- Export search results as CSV or TXT report

7.3 Quick Discovery Presets

- Find all WhatsApp media
- Find all photos (JPEG, PNG, HEIC, RAW)
- Find all videos (MP4, MOV, MKV)
- Find all documents (PDF, DOCX, PPTX, XLSX)
- Find all APK files
- Find files larger than 50 MB
- Find files older than 1 year
- Find files with no extension

Module 8: App Manager

View, manage, and clean up installed apps on the connected Android device.

8.1 App Listing

- List all installed apps (user-installed and system)
- Show per app: name, package ID, version, installed size, data size, cache size
- Filter: user apps only / system apps only / all
- Sort by: name, total size, data size, cache size, install date

8.2 Cache Management

- Clear cache for individual app
- Clear cache for all user apps in one click
- Show estimated space to be freed before clearing
- Exclude specific apps from bulk cache clear

8.3 App Uninstall

- Uninstall user-installed apps via ADB
- Batch uninstall: select multiple apps
- Confirmation dialog before uninstall
- Cannot uninstall system apps (safely blocked)

8.4 APK Extraction

- Extract APK file from any installed app to laptop
- Useful for backing up apps before uninstalling
- Show APK size before extraction

8.5 Bloatware Manager

- Identify pre-installed manufacturer apps (bloatware)
- Disable (not uninstall) bloatware apps safely
- Re-enable disabled apps
- Show which apps are currently disabled

Module 9: Reports & Analysis

Generate rich, detailed reports for all analysis and operations. Save session history for future reference.

9.1 WhatsApp Analysis Reports

- Full media inventory: all folders, all years, all months, file counts, sizes
- Pre vs post cutoff comparison report
- File type breakdown report (extension, count, total size, sorted by size)
- Sent vs Received vs Private breakdown
- Orphaned media report (files with no linked chat)
- Documents categorization report (PDFs, Archives, Images, Videos, Code, Engineering, etc.)
- Before/after cleanup comparison

9.2 Storage Reports

- Phone storage breakdown report
- Top apps by size report
- Large files report
- Storage trend: compare reports across sessions to see growth over time

9.3 Backup Reports

- Backup summary: what was backed up, file counts, sizes, duration
- Backup verification report: match status between source and destination
- Backup history log
- Session summary: all operations performed in a session

9.4 Deletion Reports

- Pre-deletion preview report: what will be deleted, broken down by year and type
- Post-deletion report: how many files deleted, how many failed, space freed
- Phone storage before vs after comparison

9.5 Report Formats

Format	Use Case
Plain Text (.txt)	Terminal-friendly, lightweight, human-readable
HTML	Rich formatted report viewable in browser
CSV	Tabular data for import into spreadsheets
JSON	Machine-readable for programmatic processing

9.6 Session Logs

- All scripts generated during a session saved to session_logs/scripts/
- Sub-categorized: analysis/, organization/, deletion/
- All reports saved to session_logs/reports/
- Session summary updated after each major operation
- Logs automatically synced to backup destination

Module 0: CLI Interface Specification

The CLI provides full access to all features via terminal commands, suitable for scripting and automation.

Command Structure

```
droidbridge <module> <command> [options]
```

Key Commands

Command	Description
droidbridge device info	Show device info and storage breakdown
droidbridge device connect	Check ADB connection and guide setup
droidbridge files browse <path>	List files at path on phone
droidbridge files search --type jpg --after 2024-01-01	Search files with filters
droidbridge transfer pull <phone_path> <local_path>	Pull files/folders from phone
droidbridge transfer push <local_path> <phone_path>	Push files/folders to phone
droidbridge whatsapp scan	Full WhatsApp media scan and report
droidbridge whatsapp analyze --cutoff 2024-09-01	Pre/post cutoff analysis
droidbridge whatsapp backup --dest /media/drive/	Full WhatsApp backup
droidbridge whatsapp backup --type voice_notes,images	Selective backup
droidbridge whatsapp organize --src /backup/ --type images	Organize backup
droidbridge whatsapp delete --before 2024-09-01 --keep images,audio	Delete with preview
droidbridge whatsapp fix-extensions --path /backup/docs/	Fix missing extensions
droidbridge storage analyze	Full phone storage analysis
droidbridge storage apps --top 20	Top apps by size
droidbridge storage large-files --min-size 50mb	Find large files

Command	Description
droidbridge backup run --profile whatsapp_full	Run backup profile
droidbridge backup verify --dest /media/drive/	Verify backup integrity
droidbridge apps list --sort size	List apps by size
droidbridge apps clear-cache --all	Clear all app caches
droidbridge report generate --format html	Generate full HTML report

Module 0: GUI Interface Specification

The GUI provides all features in a clean desktop application suitable for general users.

Layout

- Left sidebar: navigation between modules
- Top bar: device status indicator, connection info, battery
- Main area: module-specific content
- Bottom status bar: current operation progress
- Log panel (collapsible): real-time operation log

Key UI Requirements

- Dark mode and light mode support
- Responsive layout — works at 1280x720 minimum
- All destructive operations require explicit confirmation dialog
- Progress dialogs for all long-running operations with cancel button
- Color-coded status indicators (green=success, red=error, yellow=warning)
- Tooltips on all buttons and options
- Keyboard shortcuts for common actions

Module 0: Non-Functional Requirements

Requirement	Specification
Platforms	Windows 10+, Ubuntu 20.04+, macOS 12+
Android Support	Android 8.0+ (API 26+), tested up to Android 14
Transfer Speed	3–5x faster than MTP (target: 80–150 MB/s on USB 3.0)
Batch Deletion	500 files per ADB command minimum
Scan Performance	500,000+ files scanned in under 5 minutes
Memory Usage	Under 512 MB RAM for normal operations
ADB Dependency	Bundled — no user installation required
Internet Required	None — fully offline
Privacy	No telemetry, no analytics, no cloud
License	MIT Open Source License
Language	Python 3.10+ (CLI), PyQt6 or Tkinter (GUI)
Packaging	Single executable via PyInstaller per platform
Documentation	README, CLI help text, in-app tooltips
Error Handling	All ADB errors caught and reported with helpful messages
Logging	All operations logged with timestamps to session_logs/
Safety	No deletion without verified backup + explicit confirmation

Implementation Phases

The project is broken into 6 phases, each independently usable and testable. Each phase builds on the previous one without requiring rework. The CLI is fully functional after Phase 3; GUI and polish come later.

Each phase ends with a working, testable CLI tool. Do not start a phase until the previous phase's deliverables pass manual testing on a real device.

Module 1: Foundation — ADB Core & Device Manager

Goal: establish the ADB wrapper and device detection that every other module depends on.

Deliverables

- Project skeleton: droidbridge/core, modules, cli, utils, reports
- Bundled ADB binaries for Linux (Windows/macOS binaries added in Phase 4)
- core/adb.py: wrapper for adb devices, shell, pull, push, with error handling
- Module 1 — Device Manager: detection, info, storage breakdown, connection health
- CLI: droidbridge device info / connect
- Basic logging framework (session_logs/ structure)

Exit Criteria

- Can detect a connected device and print model, Android version, storage breakdown
- Handles unauthorized/offline device states gracefully with guidance
- ADB daemon restart works if connection drops

Module 2: File Operations — Browser, Search, Transfer

Goal: build the generic file operations that the WhatsApp toolkit will reuse.

Deliverables

- Module 2 — File Browser: directory listing, filtering, sorting
- Module 7 — Search & Discovery: name/type/date/size filters, quick presets
- Module 3 — Smart Transfer Engine:
 - Single-file and folder pull/push
 - Batch ADB commands (500 files per call)
 - Progress tracking (count, bytes, speed, ETA)
 - Conflict resolution (skip/overwrite/rename)
 - Post-transfer verification (count + size)
 - Resume support: detect partial transfers, skip completed files
- Sleep inhibitor for Linux (systemd-inhibit)
- CLI: droidbridge files browse / search, droidbridge transfer pull / push

Exit Criteria

- Can pull a 20+ GB folder with progress display and verify file count/size after
- Interrupting and re-running a transfer skips already-copied files
- Search returns correct results for combined filters (type + date + size)

Module 3: WhatsApp Toolkit — Analysis, Backup, Cleanup

Goal: implement the core value of the project — the full WhatsApp toolkit. This is the largest phase.

3.1 Analysis Sub-Phase

- Auto-detect WhatsApp media path (com.whatsapp and com.whatsapp.w4b)
- Full scan of all 14 media folder types
- Report by folder/section (Received/Sent/Private)/year/month/extension
- Pre/post cutoff date comparison report
- Orphaned media detection
- CLI: droidbridge whatsapp scan, droidbridge whatsapp analyze --cutoff

3.2 Backup Sub-Phase

- Full and selective backup using Phase 2 transfer engine
- Direct-to-external-drive support
- Post-backup verification (count + size match)
- Backup report + session log generation
- CLI: droidbridge whatsapp backup --dest --type

3.3 Organization Sub-Phase (Optional Step)

- Organize-by-date for Voice Notes, Images, Video (Received/Sent/Private)
- Organize-by-category for Documents (PDFs, Archives, Code, Engineering, etc.)
- MIME-type extension fixing for files with no/wrong extension
- Double-dot filename fixing
- Bad year folder detection and correction (student IDs, serial numbers)
- CLI: `droidbridge whatsapp organize --src --type`

3.4 Cleanup Sub-Phase

- Pre-deletion preview report (count + size by year/type)
- Keep/delete rules by file type and date cutoff
- Mandatory typed confirmation ('YES DELETE')
- Batch deletion (500 files per ADB call)
- Resume interrupted deletion via rescan
- Post-deletion verification and report
- Block deletion if backup not verified
- CLI: `droidbridge whatsapp delete --before --keep`

3.5 Database Sub-Phase

- Backup Databases/, Backups/, accounts/ folders
- CLI: `droidbridge whatsapp backup-db --dest`

Exit Criteria

- Full WhatsApp scan on a real device with 500k+ files completes in under 5 minutes
- Backup, organize, and delete workflow tested end-to-end on a real device
- Deletion never proceeds without a verified backup and typed confirmation
- All session logs (scripts/reports/summary) generated correctly

Module 4: Storage, Apps & Backup Manager + Cross-Platform

Goal: round out remaining modules and add Windows/macOS platform support.

Deliverables

- Module 5 — Storage Analyzer: full breakdown, top apps, large files, duplicates, suggestions
- Module 8 — App Manager: list, clear cache, uninstall, APK extraction, bloatware manager
- Module 6 — Backup Manager: profiles, incremental backup, history, restore

- Platform abstraction layer: core/platform/ (windows.py, linux.py, macos.py)
- Bundle ADB binaries for Windows and macOS
- Sleep inhibitor for Windows (SetThreadExecutionState) and macOS (caffeinate)
- Test full CLI suite on Windows and macOS

Exit Criteria

- All CLI commands work identically on Linux, Windows, and macOS
- Storage analyzer and app manager tested on a real device
- Backup profiles save/load correctly and incremental backup skips unchanged files

Module 5: Reports & Analysis Module

Goal: unify all report generation into a consistent, reusable module.

Deliverables

- Module 9 — Reports: TXT, HTML, CSV, JSON generators for all report types
- Session log management: consistent structure across all modules
- Storage trend comparison across multiple sessions
- CLI: droidbridge report generate --format

Exit Criteria

- Every module's output can be exported in all 4 formats
- HTML reports render correctly in a browser with no external dependencies

Module 6: GUI & Packaging

Goal: wrap the fully-working CLI in a desktop GUI and produce distributable executables.

Deliverables

- PyQt6 GUI with sidebar navigation matching the 9 modules
- GUI calls into the same core/ and modules/ logic as the CLI (no duplicated logic)
- Progress dialogs, confirmation dialogs, dark/light mode
- PyInstaller builds for Windows, Linux, macOS
- README, user guide, and CLI help text finalized

Exit Criteria

- GUI performs every operation the CLI can perform

- Single executable runs on a clean machine without Python installed
- All destructive operations show confirmation dialogs in GUI

Recommended approach: complete Phases 1–3 fully and use the CLI for real backups before starting Phase 4. This validates the core WhatsApp toolkit against real-world data early.

Claude Code Prompt

Use the following prompt when initiating the DroidBridge project with Claude Code CLI. It instructs Claude Code to follow the phased plan above, starting with Phase 1.

Build DroidBridge — an ADB-powered Android device management tool.

Project structure:

Create a Python project with the following layout:

```
droidbridge/  
  core/          # ADB wrapper, device management  
  modules/       # One file per module (whatsapp, storage, backup, etc.)  
  cli/           # Click-based CLI entry points  
  gui/           # PyQt6 GUI (optional, separate entry point)  
  reports/       # Report generators (TXT, HTML, CSV, JSON)  
  utils/         # Shared helpers (date parsing, size formatting, etc.)
```

Implementation approach — follow these phases in order, do not skip ahead:

PHASE 1 — Foundation: Implement core/adb.py (ADB wrapper for devices/shell/pull/push with error handling) and Module 1 (Device Manager: detection, info, storage breakdown, connection health). Add CLI commands: device info, device connect. Bundle Linux ADB binary first.

PHASE 2 — File Operations: Implement Module 2 (File Browser), Module 7 (Search & Discovery), and Module 3 (Smart Transfer Engine) with batch ADB pull/push (500 files per call), progress tracking, conflict resolution, post-transfer verification, and resume support. Add Linux sleep inhibitor (systemd-inhibit). CLI: files browse/search, transfer pull/push.

PHASE 3 — WhatsApp Toolkit (highest priority, largest phase): Implement Module 4 in 5 sub-steps — (a) Analysis: auto-detect WhatsApp/WhatsApp Business media paths, scan all 14 media folder types, report by folder/section/year/month/extension, pre/post cutoff comparison, orphaned media detection; (b) Backup: full/selective backup using Phase 2 transfer engine, direct-to-external-drive, post-backup verification, session logs; (c) Organization (OPTIONAL step, not automatic): organize-by-date for Voice Notes/Images/Video, organize-by-category for Documents, MIME-type extension fixing, double-dot filename fixing, bad year folder correction; (d) Cleanup: pre-deletion preview, keep/delete rules, mandatory typed 'YES DELETE' confirmation, batch deletion (500/call), resume via rescan, post-deletion verification, block deletion if backup unverified; (e) Database: backup Databases/, Backups/, accounts/ folders. CLI: whatsapp scan/analyze/backup/organize/delete/backup-db.

PHASE 4 — Storage, Apps, Backup Manager + Cross-Platform: Implement Module 5 (Storage Analyzer), Module 8 (App Manager), Module 6 (Backup Manager with profiles/incremental/restore). Add core/platform/ abstraction layer (windows.py, linux.py, macos.py), bundle Windows and macOS ADB binaries, add Windows/macOS sleep inhibitors. Test full CLI on all 3 platforms.

PHASE 5 — Reports & Analysis: Implement Module 9 with TXT/HTML/CSV/JSON report generators reused by all modules, unify session log structure, add storage trend comparison across sessions. CLI: report generate --format.

PHASE 6 — GUI & Packaging: Build PyQt6 GUI with sidebar navigation for all 9 modules, calling the same core/modules logic as CLI (no duplication). Add progress/confirmation dialogs, dark/light mode. Produce PyInstaller executables for Windows, Linux, macOS.

Key technical requirements:

- Single Python codebase shared across platforms; platform differences isolated in core/platform/
- Bundle ADB platform-tools per platform (Linux first in Phase 1, Windows/macOS in Phase 4)
- Backup organization (by date for media, by category for documents) is OPTIONAL — a separate user-triggered step, never automatic
- All destructive operations require verified backup + explicit typed 'YES DELETE' confirmation
- Batch ADB commands (500 files per rm call) for fast deletion
- Sleep inhibitor during long transfers (systemd-inhibit on Linux; add Windows/macOS in Phase 4)
- WhatsApp path auto-detection for both com.whatsapp and com.whatsapp.w4b (supports Android 8–14)
- Date validation: year 2009–2030, month 01–12, reject false matches from serial numbers/student IDs
- MIME type detection for files with missing/wrong extensions
- All operations resumable after interruption (transfers and deletions)
- Fully offline — no internet, no cloud, no telemetry
- Open source, MIT license

Start with Phase 1 right now:

1. Create the project structure and requirements.txt (click, etc.)
2. Implement core/adb.py — the ADB wrapper with devices/shell/pull/push and error handling
3. Implement modules/device.py — Module 1 (Device Manager)
4. Implement cli/main.py — Click-based CLI entry point with 'device info' and 'device connect' commands
5. Test against a real connected Android device before moving to Phase 2

This document was generated as part of the DroidBridge open source project specification. Start with the WhatsApp Toolkit module as it contains the most complex and battle-tested logic.